# Proving Data Structures Correct: Balanced Search Trees

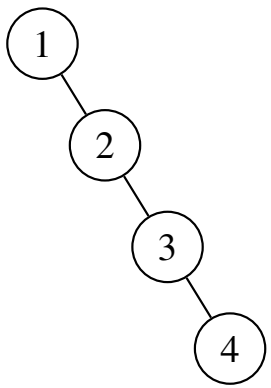
Computer-Assisted Theorem Proving (7CCSACTP)

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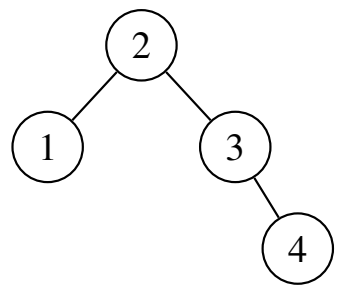
# Search Time Depends on the Height

Week 4: `isin` follows one path from the root
$\Rightarrow$ search takes time $O(h(t))$, where $h(t)$ is the height of $t$

The height depends on the order of insertion:




`ins` 1, 2, 3, 4: height 4




`ins` 2, 1, 3, 4: height 3

Worst case: $h(t) = |t|$

Best case: $h(t) \approx \log_2 |t|$

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# Balanced Search Trees

*Balanced* search trees guarantee $h(t) = O(\log |t|)$

Two kinds of invariants:

- the ordering invariant, like `bst` in Week 4
- a structural invariant = a *balance criterion*

`insert` (and `delete`) must preserve both.

Examples: AVL trees, red-black trees, 2-3 trees, B-trees, ...

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